Predictors of High-Cost Patient Status Among U.S. Adults: Logistic Regression vs. Naive Bayes Using MEPS 2024 Data

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Abstract

Healthcare spending in the United States is concentrated in a small share of patients. This study used 2024 Medical Expenditure Panel Survey (MEPS) data on 14,764 adults, defining high-cost patients as the top 10% of total spending. It tested whether insurance status was associated with the odds of being high-cost after adjusting for age, sex, race/ethnicity, poverty level, chronic conditions, and self-rated health, and it compared logistic regression with Naive Bayes for identifying high-cost adults. In survey-weighted logistic regression, uninsured adults had lower odds than privately insured adults (OR = 0.135, 95% CI 0.060 to 0.304). Logistic regression discriminated slightly better than Naive Bayes on a household-grouped test set (ROC-AUC 0.807 vs. 0.790) and in cross-validation (0.782 vs. 0.755), though both performed only moderately. Basic demographic, insurance, and health characteristics therefore carry some information about high-cost status but cannot pinpoint individual high-cost patients. The cross-sectional design precludes causal conclusions, and only nine high-cost adults were uninsured, so the insurance estimate is imprecise.

1. Introduction

Health care spending in the United States is highly concentrated. A small share of the population accounts for a disproportionately large share of total expenditures, while the majority of people account for relatively little (Cohen, 2012). This concentration makes prediction valuable. Insurers and hospitals have strong incentives to identify likely high-cost patients early, before costs accumulate, so they can target care management, coordinate services, and allocate resources. Accurate prediction from data that are easy to collect makes such targeting more practical.

Insurance coverage has long been linked to health care use and costs. Evidence indicates that insured and uninsured individuals use and spend different amounts on health care services (Hadley et al., 2008). Coverage status is therefore a plausible predictor of high-cost status.

This study uses a small set of variables that are easy to collect: age, sex, race and ethnicity, poverty level, insurance status, number of chronic conditions, and self-rated physical and mental health. The data come from the 2024 Medical Expenditure Panel Survey (MEPS), a nationally representative survey of U.S. adults. The study combines a formal test of the insurance association with a comparison of two classifiers, logistic regression and Naive Bayes, on how well they identify the top 10% of spenders.

Two research questions guide the analysis. First, after adjusting for demographics, poverty level, chronic conditions, and self-rated health, are uninsured adults less likely than privately insured adults to be high-cost patients? Second, how well can these characteristics identify high-cost patients, and does logistic regression or Naive Bayes perform better? The null hypothesis (H0) is that the odds of being a high-cost patient do not differ between uninsured and privately insured

adults (OR = 1). The alternative hypothesis (H1) is that uninsured adults have lower odds (OR < 1).

The rest of the paper describes the data and methods, presents exploratory and model results, and interprets the findings and their limits.

2. Literature Review

Healthcare spending in the United States is concentrated in a small group of people. A very small percentage of the population accounts for most healthcare costs, while most people spend very little (Cohen, 2012). Focusing on spending in this way is useful for predicting who will be expensive to care for. Insurers and hospitals want to identify high-cost patients proactively so they can manage care, coordinate services, and use resources effectively, and this is much more practical with simple data that are easy to collect.

Health insurance affects how often people use healthcare and how much they spend. Adults without insurance generally use less care than those with insurance (Hadley et al., 2008). However, insured and uninsured adults also have different health needs, so comparing their spending without accounting for health status can be misleading. To see whether insurance status is associated with high healthcare spending, differences in overall health must be taken into account.

Health needs have been studied directly as key drivers of healthcare spending. Research shows that spending increases as the number of chronic conditions a person has grows (Meraya et al., 2015). A single question about self-rated health is an easy-to-use measure. A review of 27 community studies found that it was consistently associated with mortality, even when other health risk factors were taken into account (Idler & Benyamini, 1997). Self-rated health is therefore well established as an overall summary indicator of an individual's health.

Predicting high-cost patients has increasingly relied on machine learning techniques. For instance, Zhang et al. (2025) used a seven-year cohort dataset to predict high-need, high-cost pediatric patients in China, evaluating several algorithms to determine the best model.

Few studies of this kind combine a formal test of the insurance association with a comparison of simple classifiers on a small set of survey variables. This study addresses that gap. It uses a small set of survey variables from a nationally representative sample of U.S. adults to test whether insurance status is associated with being a high-cost patient after controlling for overall health. It also compares two predictive models: logistic regression and Naive Bayes. Based on past literature, we hypothesize that uninsured adults have lower odds of being high-cost patients than privately insured adults.

3. Data

Data source. The information comes from the Household Component of the Medical Expenditure Panel Survey (MEPS), 2024 Full-Year Consolidated Data File (HC-256), provided by the Agency for Healthcare Research and Quality (AHRQ): https://meps.ahrq.gov/mepsweb/data_stats/download_data_files_detail.jsp?cboPUFNumber=HC-256. MEPS is a nationally representative survey of the U.S. civilian noninstitutionalized population and is frequently used in health economics. It was selected because it contains health care expenditures at the individual level along with insurance status, health, and demographic characteristics, and it is available in a single public file.

Sample. There are 19,140 persons in the file. The analysis was limited to adults (aged 18 or older) who had a positive person-level survey weight (15,126 adults), because the chronic condition questions are asked only of adults. Another 362 adults were excluded because of missing information on one or more predictors, resulting in a sample of 14,764 adults from 8,931 households (an average of 1.65 adults per household).

Outcome. The outcome is total health care expenditure in 2024 (TOTEXP24). Spending was very skewed, with a median of $2,850, a mean of $10,800, and a maximum of $848,000 for the highest-spending adult; 12.5% of adults reported no spending at all. High-cost patients were defined as adults in the top 10% of spending, which corresponded to a threshold of $26,379. Adults with no spending were retained in the non-high-cost group.

Predictors and cleaning. Predictors were age, sex, race and ethnicity, insurance status, poverty category, number of chronic conditions, and self-rated physical and mental health (Table 1). The chronic condition count is the number of nine diagnosed conditions reported: high blood pressure, heart disease, stroke, emphysema, high cholesterol, cancer, diabetes, arthritis, and asthma. MEPS codes with a negative value were set to missing before analysis. The extreme spenders were purposely retained because they are the target population of the study.

Table 1. Predictors and coding

Predictor	MEPS variable	Coding	Reference group
Age	AGELAST	Years (18 to 85)	-
Sex	SEX	Female = 1	Male
Race/ethnicity	RACETHX	4 dummy columns	Non-Hispanic White
Insurance	INSCOV24	2 dummy columns	Any private
Poverty level	POVCAT24	4 dummy columns	High income
Chronic conditions	9 diagnosis flags	Count, 0 to 9	-
Physical health	RTHLTH53	1 (excellent) to 5 (poor)	-
Mental health	MNHLTH53	1 (excellent) to 5 (poor)	-

Table 2. Summary statistics of the 15 predictor columns (n = 14,764)

Variable	Mean	SD	Min	Max
age	52.847	18.502	18.0	85.0
female	0.530	0.499	0.0	1.0
race_hispanic	0.194	0.395	0.0	1.0
race_black	0.125	0.331	0.0	1.0
race_asian	0.057	0.232	0.0	1.0
race_other	0.031	0.173	0.0	1.0
ins_public	0.335	0.472	0.0	1.0
ins_uninsured	0.070	0.256	0.0	1.0
pov_poor	0.142	0.349	0.0	1.0
pov_near_poor	0.041	0.197	0.0	1.0
pov_low	0.115	0.319	0.0	1.0
pov_middle	0.266	0.442	0.0	1.0
chronic_count	1.635	1.639	0.0	9.0
phys_health	2.486	1.023	1.0	5.0
mental_health	2.344	1.015	1.0	5.0

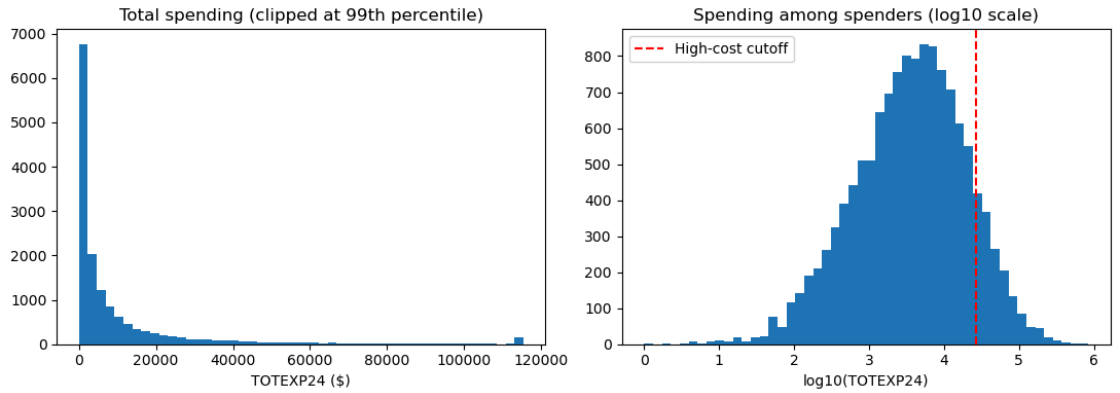


Figure 1. Distribution of total spending (left, clipped at the 99th percentile) and of spending among spenders on a log10 scale (right). The dashed line marks the high-cost cutoff.

4. Methods

Hypothesis and outcome. The first objective was to determine whether insurance status is linked to high-cost patient status. The null hypothesis (H0) was that, after adjusting for the other predictors, the odds of being a high-cost patient are equal (OR = 1) for uninsured and privately insured adults. The alternative hypothesis (H1) was that uninsured adults have lower odds (OR < 1). High-cost patients were adults in the top 10% of total expenditures (approximately $26,379 in 2024). The cutoff was determined without survey weights.

Logistic regression. All 15 predictor columns (the eight predictors, with categorical variables coded as indicator columns against the reference groups listed in Table 1) were used in the logistic regression model to test the hypothesis. The model was fitted on the entire analysis sample using the person-level survey weight (PERWT24F, scaled to a mean of 1), with standard errors clustered by household because adults within a household are not independent. Odds ratios and 95% confidence intervals are reported. Because H1 is directional, the uninsured coefficient was tested one-sided at the 0.05 level. As a sensitivity check, the model was refitted without survey weights.

Assumption checks. Before modelling, we checked that there were enough events per predictor (at least 10 as a rule of thumb), tested linearity of the logit for the two numeric predictors (age and number of chronic conditions) with the Box-Tidwell test, and examined variance inflation factors (VIF) for multicollinearity. We also examined the household structure, which motivated clustering the standard errors and splitting the data by household, and the correlations between predictors, which matter for Naive Bayes.

Prediction models. The second objective was to evaluate how well two classifiers identify high-cost adults. Logistic regression was fitted on standardized predictors with no regularization penalty, and Gaussian Naive Bayes was fitted on the unstandardized predictors. The two were chosen because they rest on different assumptions: logistic regression estimates the effect of each predictor while holding the others fixed, whereas Naive Bayes treats each predictor as independent of the others given the outcome. Comparing them shows the cost of the independence assumption when predictors are correlated. Survey weights were not used in the prediction models.

Training and testing. To prevent information from leaking between the training and test data, the data were divided into a training set (11,751 adults) and a test set (3,013 adults), with all adults in a household assigned to the same set. We also ran five-fold cross-validation grouped by household and stratified by the outcome, so that each fold contained a similar proportion of high-cost adults. The random seed was 42.

Evaluation. The models were evaluated on the test set using the area under the ROC curve (ROC-AUC), the area under the precision-recall curve (PR-AUC), precision, recall, the F1 score, the Brier score, and confusion matrices. Accuracy was not used as a primary measure because only about 10% of adults are high-cost, so a model that classifies every adult as “not high-cost” would be about 91% accurate. Precision, recall, and F1 were calculated for two decision rules: a predicted probability of at least 0.5, and flagging the 10% of adults with the highest predicted probability. The second rule allows a fair comparison when the models produce probabilities on different scales.

Software. Analyses were run in Python using pandas, NumPy, statsmodels, scikit-learn, and matplotlib.

5. Results

5.1 Exploratory Data Analysis: Who the High-Cost Patients Are

High-cost adults were older (mean 62.6 vs. 51.8 years) and had about twice as many chronic conditions (3.00 vs. 1.48), and almost none were uninsured (0.6% vs. 7.7%) (Table 3). The share of adults who were high-cost rose steadily with the number of chronic conditions, from about 3% with none to about 27% with four or more (Figure 2). Figure 3 shows the correlations between predictors. These are unadjusted comparisons, so the regression analysis is needed to separate the effects.

Table 3. Predictor means or shares by high-cost status

Variable (mean or share)	Not high-cost	High-cost
Age (years)	51.8	62.6
Female	0.525	0.575
Hispanic	0.204	0.107
Black	0.127	0.103
Asian	0.060	0.029
Other race	0.031	0.030
Public insurance	0.322	0.447
Uninsured	0.077	0.006
Poverty: poor	0.140	0.158
Poverty: near poor	0.040	0.045
Poverty: low income	0.114	0.121
Poverty: middle income	0.269	0.238
Chronic conditions	1.48	3.00
Physical health rating (1 to 5)	2.41	3.15
Mental health rating (1 to 5)	2.31	2.67

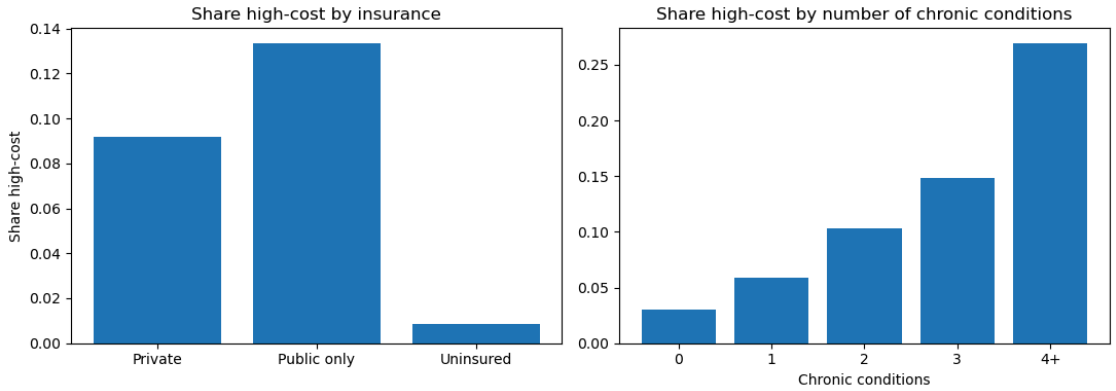


Figure 2. Share of adults who were high-cost, by insurance type (left) and by number of chronic conditions (right).

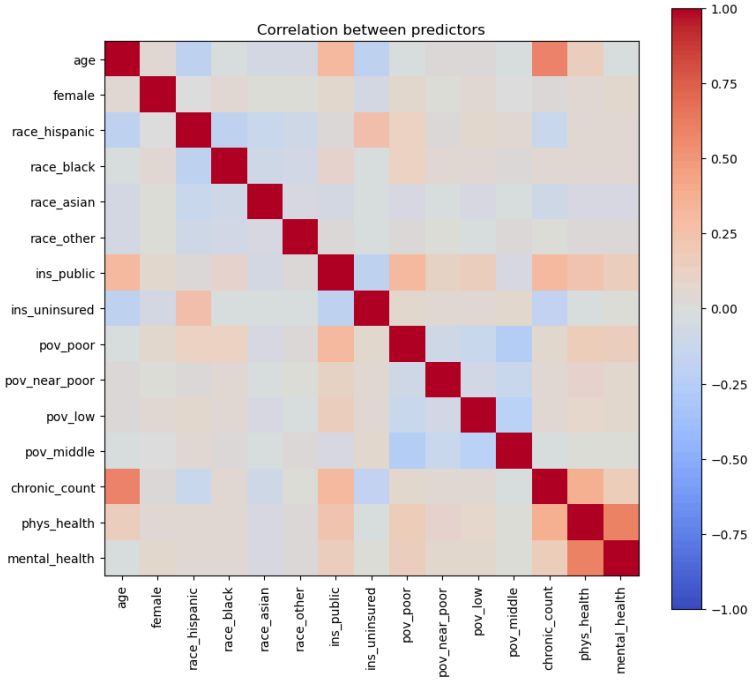


Figure 3. Correlation heatmap of the predictors.

5.2 Assumption Testing

Table 4. Assumption checks

Check	Result
Events per predictor	1,477 high-cost adults, 98 per predictor (rule of thumb: 10)
Multicollinearity (VIF)	All between 1.02 and 1.83
Box-Tidwell, age	p = 0.295 (no evidence of non-linearity)
Box-Tidwell, chronic conditions	p = 0.319 (no evidence of non-linearity)
Independence	1.65 adults per household, so clustering and grouped splits were used

Check	Result
Strongest predictor correlations	Physical and mental health 0.607; age and chronic conditions 0.587; chronic conditions and physical health 0.370

5.3 Hypothesis Test (H1)

Table 5. Test of H1: uninsured vs. privately insured adults

Item	Value
Uninsured coefficient	-2.00
Odds ratio (95% CI)	0.135 (0.060 to 0.304)
Two-sided p	< 0.001
One-sided p (H1: OR < 1)	< 0.001
Decision	Reject H0
Unweighted check	OR about 0.16, p < 0.001
High-cost uninsured adults	9 of 1,038 uninsured adults

After adjusting for age, sex, race and ethnicity, poverty level, chronic conditions, and self-rated health, uninsured adults had substantially lower odds of being high-cost than privately insured adults (OR = 0.135, 95% CI 0.060 to 0.304; one-sided p < 0.001), so the null hypothesis was rejected. This corresponds to odds about 86% lower. The result was similar without survey weights (OR = 0.16, p < 0.001). The confidence interval is wide because only 9 of the 1,038 uninsured adults were high-cost.

5.4 All Predictors (Odds Ratios)

Table 6. Adjusted odds ratios for being a high-cost patient

Predictor	Odds ratio	95% CI	p-value	Clear effect?
Age (per year)	1.014	1.009 to 1.020	<0.001	Yes
Female	1.288	1.124 to 1.475	0.0003	Yes
Hispanic	0.606	0.480 to 0.766	<0.001	Yes
Black	0.579	0.457 to 0.732	<0.001	Yes
Asian	0.588	0.401 to 0.861	0.006	Yes
Other race	0.775	0.483 to 1.243	0.290	No
Public insurance	0.785	0.668 to 0.922	0.003	Yes
Uninsured	0.135	0.060 to 0.304	<0.001	Yes
Poverty: poor	0.824	0.647 to 1.050	0.117	No
Poverty: near poor	0.761	0.512 to 1.132	0.178	No
Poverty: low income	0.742	0.597 to 0.922	0.007	Yes

Predictor	Odds ratio	95% CI	p-value	Clear effect?
Poverty: middle income	0.779	0.655 to 0.927	0.005	Yes
Chronic conditions (each)	1.354	1.291 to 1.420	<0.001	Yes
Physical health (per step worse)	1.773	1.617 to 1.944	<0.001	Yes
Mental health (per step worse)	0.984	0.904 to 1.071	0.707	No

Table 6 shows the adjusted odds ratios; the reference groups are private insurance, non-Hispanic White, high income, and male. Worse self-rated physical health had the largest association with being high-cost (OR = 1.77 per step on the 5-point scale, 95% CI 1.62 to 1.94), followed by the number of chronic conditions (OR = 1.35 per condition, 95% CI 1.29 to 1.42), and being uninsured was associated with much lower odds than private coverage (OR = 0.135). Odds were also higher with each additional year of age (OR = 1.014) and for women (OR = 1.29), and lower for adults with public insurance (OR = 0.79). Hispanic, Black, and Asian adults had lower odds than non-Hispanic White adults (ORs of about 0.58 to 0.61), as did adults in the low-income and middle-income categories compared with high-income adults (ORs of 0.74 and 0.78). Self-rated mental health was not associated with high-cost status once the other variables were included (OR = 0.98, 95% CI 0.90 to 1.07, p = 0.707), and neither were the poor and near-poor categories or the “other race” group, whose confidence intervals included 1 (Figure 4).

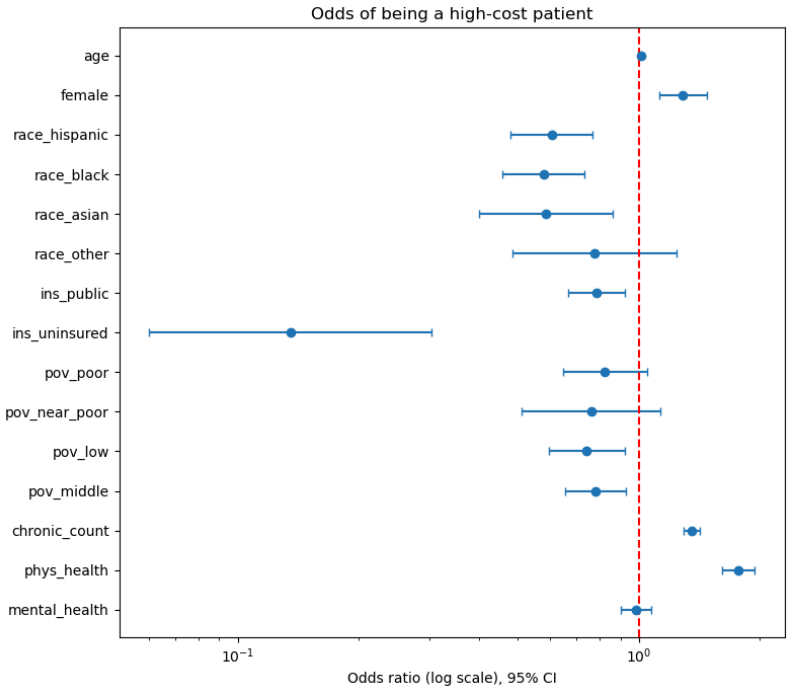


Figure 4. Adjusted odds ratios (log scale) with 95% confidence intervals.

5.5 Prediction Results

The test set contained 3,013 adults, 270 of whom (9.0%) were truly high-cost.

Table 7. Test-set performance of logistic regression and Naive Bayes

Score	Logistic	Naive Bayes
ROC-AUC	0.807	0.790
PR-AUC	0.304	0.303
Brier score (lower is better)	0.071	0.156
Precision at 0.5 cutoff	0.520	0.230
Recall at 0.5 cutoff	0.048	0.637
F1 at 0.5 cutoff	0.088	0.338
Precision, top 10% flagged	0.344	0.308
Recall, top 10% flagged	0.385	0.344
Cross-validated AUC, mean (SD)	0.782 (0.006)	0.755 (0.012)

Table 8. Confusion matrices at the 0.5 cutoff

Confusion matrix at 0.5	Predicted no	Predicted yes
Logistic: actual no	2,731	12
Logistic: actual yes	257	13
Naive Bayes: actual no	2,166	577
Naive Bayes: actual yes	98	172

Both classifiers ranked high-cost adults well above chance on the held-out test set of 3,013 adults, 270 of whom were high-cost (Table 7). Logistic regression had a slightly higher ROC-AUC than Naive Bayes (0.807 vs. 0.790), and the two had nearly identical PR-AUC (0.304 vs. 0.303), against a random-guessing baseline of about 0.09 (Figure 5). The same ordering held in five-fold cross-validation (mean AUC 0.782 vs. 0.755). When each model flagged the 10% of adults with the highest predicted risk, logistic regression reached a precision of 0.344 and a recall of 0.385, compared with 0.308 and 0.344 for Naive Bayes.

The models behaved very differently at the 0.5 probability cutoff (Table 8). Logistic regression flagged only 25 adults as high-cost, 13 of them correctly, and so missed 257 of the 270 high-cost adults (recall 0.048). Because only about 9% of adults are high-cost, its predicted probabilities rarely reach 0.5. Naive Bayes flagged 749 adults and identified 172 of the 270 (recall 0.637), but 577 of its flags were false alarms (precision 0.230). Naive Bayes also had a much higher Brier score (0.156 vs. 0.071), meaning its predicted probabilities were less well calibrated. Accuracy was not informative, since classifying every adult as not high-cost would be about 91% accurate.

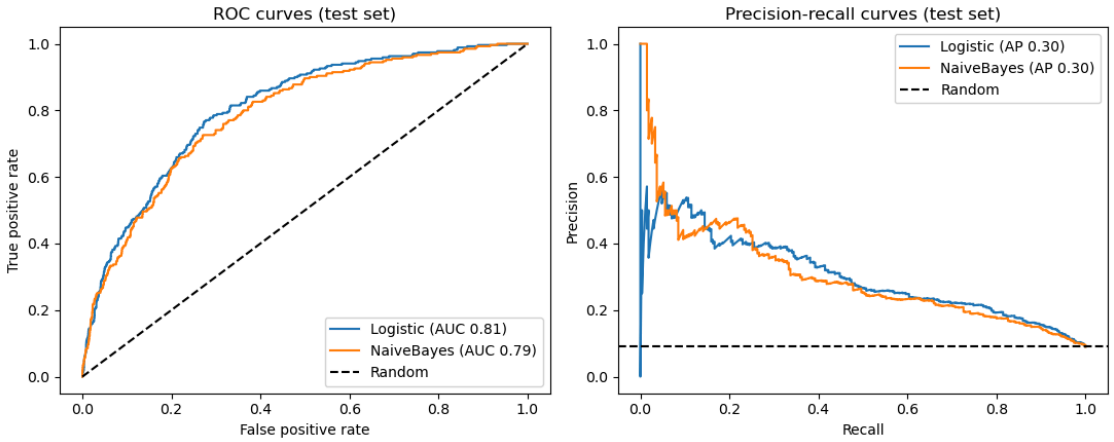


Figure 5. ROC curves (left) and precision-recall curves (right) on the test set.

6. Discussion and Interpretation

Uninsured adults have lower recorded costs. Uninsured adults were significantly less likely to be classified as high-cost than privately insured adults of similar age, background, and health status. One possible reason is that uninsured people often skip or delay necessary medical care, so their spending is lower. Another is that people who develop extremely costly conditions may qualify for public coverage such as Medicaid, which would move them out of the uninsured group. This study cannot test either explanation. The estimate also carries a wide margin of error, as only 9 of the 1,038 uninsured adults in the sample were high-cost, though the odds remain consistently lower than for insured adults.

Public vs. private insurance. At first glance, adults with public insurance appear to generate higher medical costs than those with private insurance (Table 3). However, this raw comparison is misleading because public programs enroll older and sicker people. Once the data are adjusted to compare public and private coverage among adults with the same age, health ratings, and chronic conditions, publicly insured adults actually have slightly lower odds of being high-cost (OR = 0.79).

Predicting high-cost patients. The models showed moderate ability to distinguish high-cost adults from lower-cost ones: with a ROC-AUC of 0.807, logistic regression ranks a randomly chosen high-cost adult above a randomly chosen other adult about 80% of the time. Even so, precision was low. When the model flagged the top 10% of adults by predicted risk, only about 34% of those flagged were actually high-cost, and only about 38% of all high-cost adults were caught. Basic survey variables such as demographics and health status provide useful context, but they cannot reliably predict individual spending, probably because costs also depend on sudden, unpredictable events such as accidents or new diagnoses.

Logistic regression vs. Naive Bayes. Logistic regression slightly outperformed Naive Bayes on ROC-AUC and in cross-validation, while the two were nearly tied on PR-AUC. A likely reason

is that Naive Bayes assumes the predictors are independent given the outcome, and this assumption is not met here: age, chronic conditions, and physical health are moderately correlated (correlations of about 0.37 to 0.61), so Naive Bayes may count the same underlying information more than once. Naive Bayes also had poorer probability calibration (Brier score 0.156 vs. 0.071), and its Gaussian form may be a poor fit for yes/no predictors.

Mental health and healthcare costs. After controlling for physical health and chronic conditions, self-rated mental health showed no independent association with high healthcare costs. This may be because self-rated physical and mental health overlap (correlation 0.607), so physical health and chronic conditions already capture most of the shared effect. These findings reflect basic self-ratings rather than formal clinical diagnoses and should not be interpreted as proof that mental health conditions are unrelated to healthcare spending.

7. Conclusion

Uninsured adults had much lower odds of being high-cost than privately insured adults after adjusting for health and demographics, and logistic regression discriminated slightly better than Naive Bayes (test AUC 0.807 vs. 0.790). Simple survey characteristics carry useful but only moderate information about who becomes a high-cost patient, so they can support early targeting but not pinpoint individual patients. These findings are associations from one year of cross-sectional, self-reported data, and future work with longitudinal data could clarify the role of insurance.

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